Danantara Indonesia

Pegadaian

Data Driven Cross Selling Berbasis Machine Learning untuk Bisnis Pegadaian



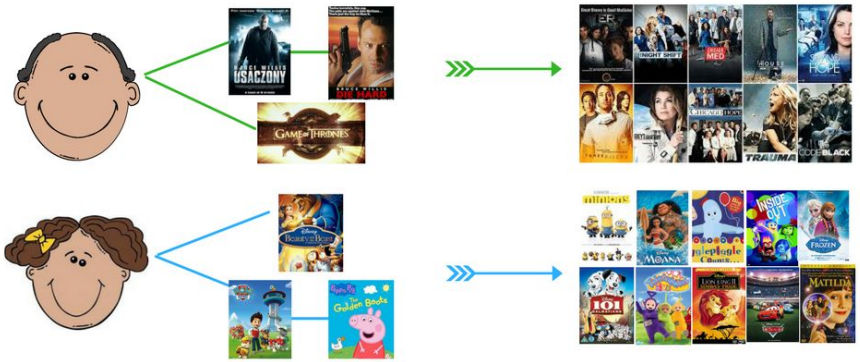
Divisi Manajemen Data TI

(Restricted, do not share without permission)

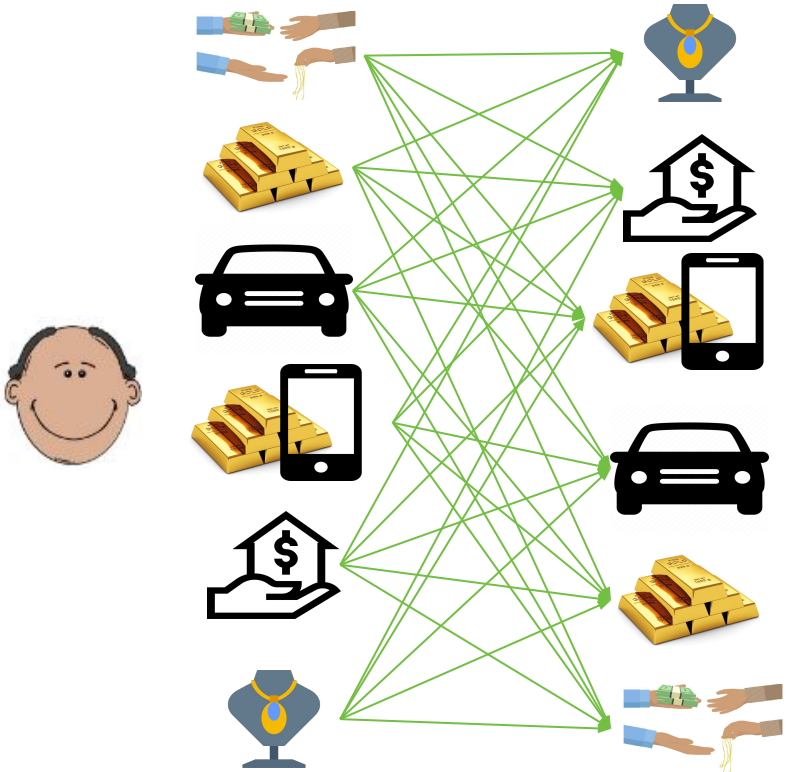
29 September 2026

Sebuah Analogi

Rekomendasi Film



Rekomendasi Cross Selling



Identifikasi Masalah Cross Selling

<= Tahun 2020

Tenaga Penjualan

Intuisi, Subyektif, Berdasar Pengalaman

Penawaran Product ke Nasabah Eksisting

Tidak sesuai preferensi nasabah, Belum berani memasang target

Tahun 2020 - 2021

Tenaga Penjualan

Intuisi & Pengalaman

+

Association Rule Mining

Penawaran ke Nasabah

Monitoring dan Evaluasi

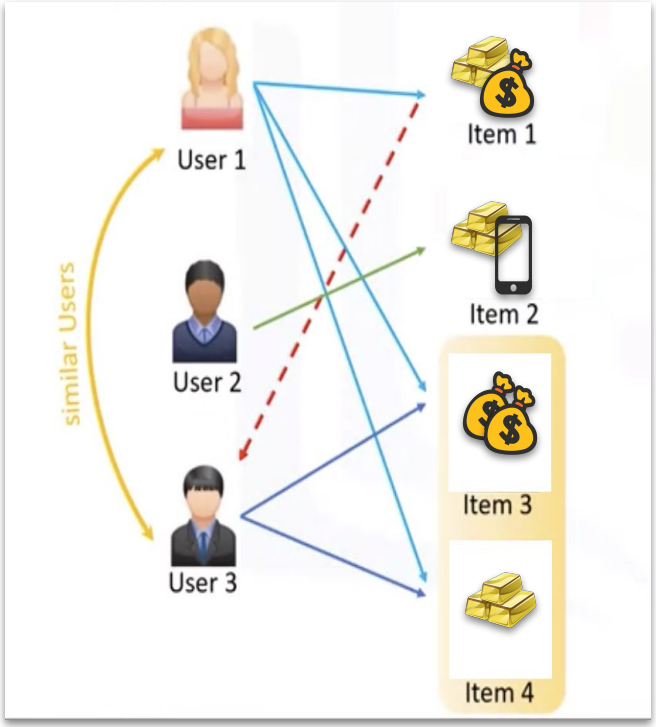


Closing Rate 17%, Akurasi masih rendah (4%)

Tahun 2022

Smart Adaptive Recommendation

Solusi Awal: Association Rules



$$Rule: X \rightarrow Y$$

$$Support(X \rightarrow Y) = \frac{freq(X,Y)}{N}$$

$$Confidence(X \rightarrow Y) = \frac{freq(X,Y)}{freq(X)}$$

$$Lift(X \rightarrow Y) = \frac{Support(X \cup Y)}{Support(X) \times Support(Y)}$$

Legend:
$freq(A)$: Frekuensi kemunculan itemset A
$Support(A)$: Probabilitas kemunculan itemset A $(= freq(A)/N)$
$X \cup Y$: Gabungan itemset X dan Y
N: Total jumlah transaksi

- Hanya memperhitungkan frekuensi dibelinya produk-produk secara bersamaan
- Belum memperhitungkan data yang menggambarkan karakter pembeli produk

Collaborative Filtering Sebagai Ide Dasar Metode Smart Adaptive Recommendation System (SAR)

Negative Interactions for Improved Collaborative Filtering: Don't go Deeper, go Higher

Machine Learning | Recommendations | Personalization & Search

Publication

September 27, 2021

Abstract

The recommendation-accuracy of collaborative filtering approaches is typically improved when taking into account higher-order interactions. While deep nonlinear models are theoretically able to learn higher-order interactions, their capabilities were, however, found to be quite limited in practice. Moreover, the use of low-dimensional embeddings in deep networks may severely limit their expressiveness. This motivated us in this paper to explore a simple extension of linear full-rank models that allow for higher-order interactions as additional explicit input-features. Interestingly, we observed that this model-class obtained by far the best ranking accuracies on the largest data set in our experiments, while it was still competitive with various state-of-the-art deep-learning models on the smaller data sets. Moreover, our approach can also be interpreted as a simple yet effective improvement of the (linear) HOSLIM model: by simply removing the constraint that the learned higher-order interactions have to be non-negative, we observed that the accuracy-gains due to higher-order interactions more than doubled in our experiments. The reason for this large improvement was that large positive higher-order interactions (as used in HOSLIM) are relatively infrequent compared to the number of large negative higher-order interactions in the three well-known data-sets used in our experiments. We further characterize the circumstances where the higher-order interactions provide the most significant improvements.

Read More

research.netflix.com/publication/



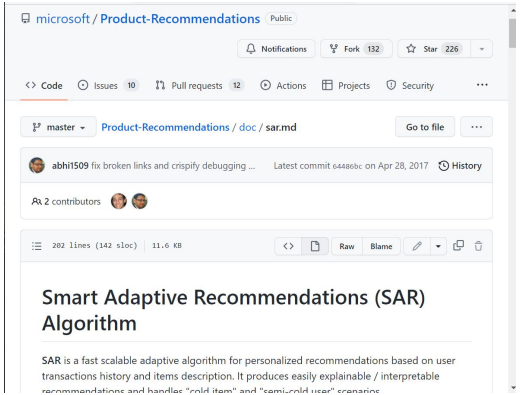
Recommender systems are machine learning algorithms programmed to suggest relevant items to specific users.[i] They are extremely valuable to many industries and generate a huge amount of income when they are efficient. Recommender systems are at the heart of many ecommerce websites such as Netflix, Amazon and LinkedIn. Data from Netflix's site shows that 80% of what people watch comes from their recommendation algorithm[ii]. As a result, recommender systems generate over $1 billion per year of Netflix's revenue.

To emphasise how important and valuable recommender systems are to a tech business, in 2006, Netflix organised a competition with a prize of $1 million, challenging the computer science community to produce a recommender system better than their own.[iii]

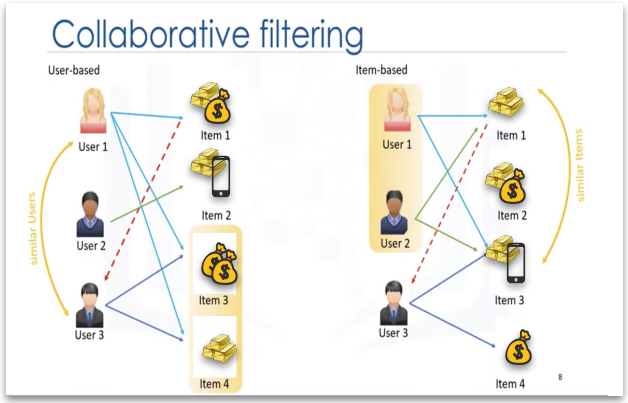
You may be familiar with that lazy Sunday (or any day – who's watching ;)), lying back on your couch, turning on the TV and scrolling through Netflix's "Top Picks for you", searching for the next best show to binge watch. What you may not be familiar with is the inner workings of the algorithms and data structures hidden behind that simple user interface. The two most commonly used recommender systems are content-based filtering and collaborative filtering. In this post, we will focus on collaborative filtering as this is used by Netflix to make our Sundays more enjoyable.

https://reflect.ucl.ac.uk/nsci0010-2021-class-blog/2021/02/24/how-netflix-uses-recommender-systems/

Microsoft



Collaborative Filtering Sebagai Ide Dasar Metode Smart Adaptive Recommendation System (SAR)

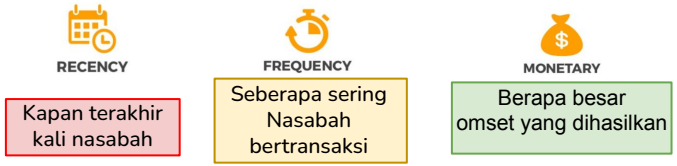


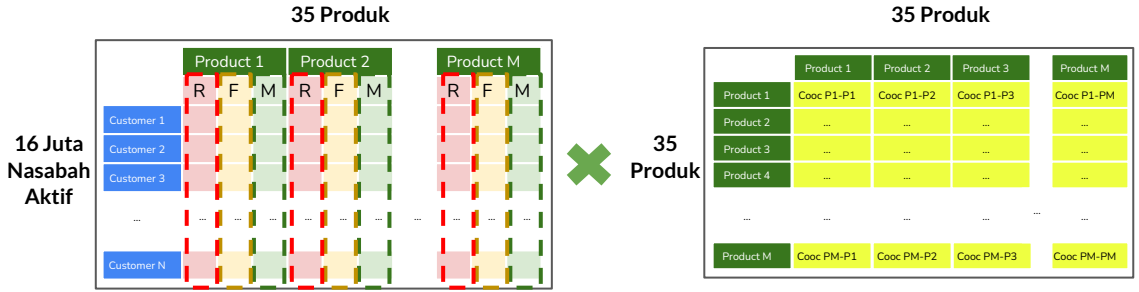
Ide dasar model SAR

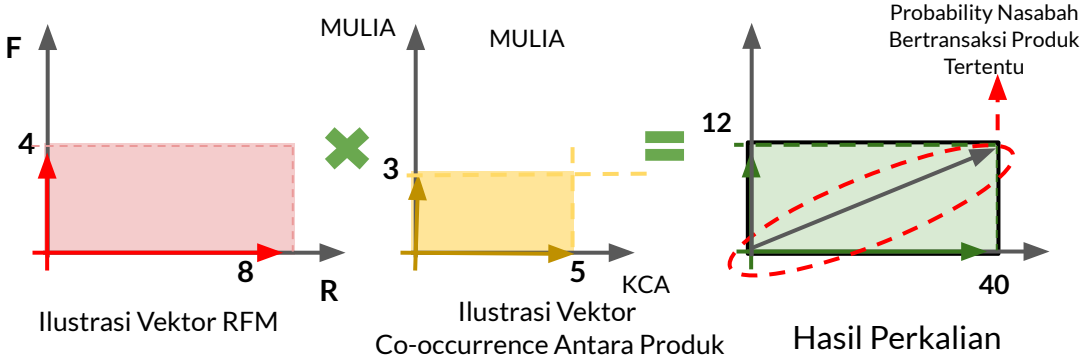
Notable Highlights:

1. Perkalian matrix yang sangat besar (16 juta x 35)
2. Eksekusi di mesin big data
3. Implementasi Formula Inverse Product Frequency (IPF)

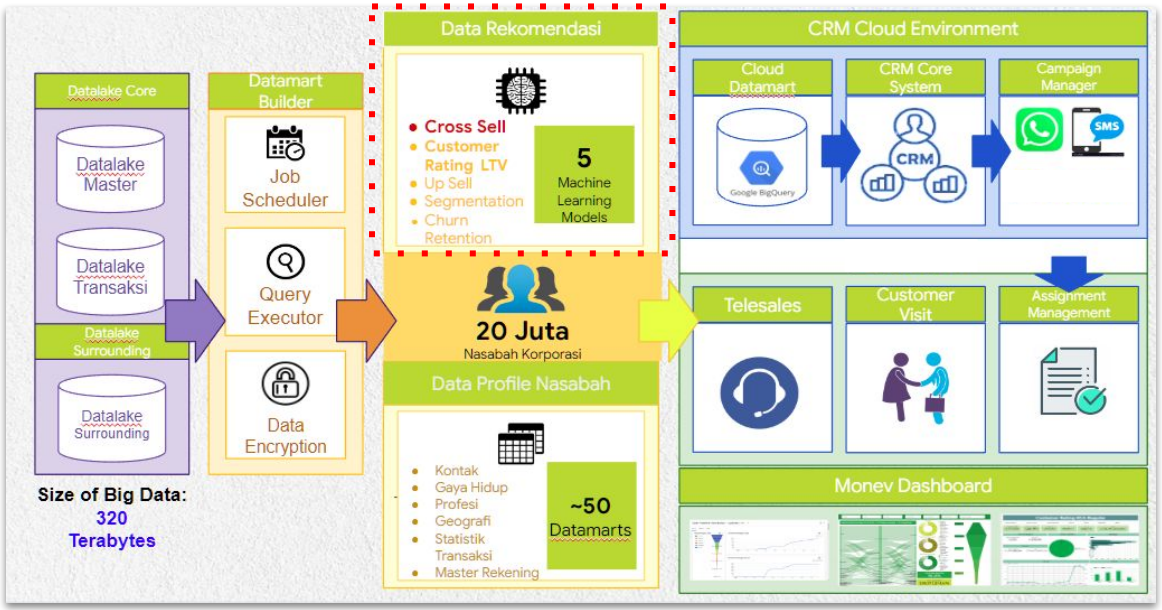
$$ipf_i = log\left(\frac{n}{pf_i}\right)$$







Integrasi Cross Selling SAR dengan CRM



Model Cross Selling dengan algoritma SAR dikembangkan dengan tahapan pembuatan datamart menggunakan query executor, dan pengamanan data pribadi dengan modul enkripsi yang ada di platform Big Data.

Model dijalankan setiap minimal dua minggu sekali, kemudian hasil rekomendasi produk dikirimkan ke ekosistem Customer Relationship Management (CRM). Di ekosistem ini, tim penjualan kemudian melakukan pendekatan kepada nasabah, menawarkan produk yang belum dimiliki berdasarkan rekomendasi. Terdapat beberapa cara untuk menjangkau nasabah mulai dari telesales, pengiriman SMS/WA Blast, sampai dengan visitasi.

Hasil tindak lanjut di lapangan ini kemudian dievaluasi secara berkala di platform Big Data untuk memonitor kualitas dan akurasi dari rekomendasi.

Thank You!